

LAI, HSIN HUI

Frontend Engineer

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SUMMARY

Frontend Engineer with 5+ years of experience building React and Next.js products, with a focus on frontend architecture, cross-platform integration, and developer productivity.

EXPERIENCE

ViewSonic

Frontend Engineer | Nov 2024 - Present

- Designed and implemented a **WebView-to-Windows architecture** for quiz document export, defining a minimal bridge contract and separating document generation, preview, and native file storage.
- Refactored quiz and task state management using **Redux and RTK Query**, separating shared state and data access from feature modules.
- Implemented classroom **game score synchronization** using a score queue and student state management.
- Led the development of a shared **AI-assisted workflow** for frontend and backend engineers, standardizing feature specs and task-specific context through **OpenSpec** and a project knowledge base.

AlfredCamera

Frontend Engineer | Nov 2022 - Oct 2024

- Rebuilt the entire official website with **Next.js**, achieving a desktop **PageSpeed Insights performance score of 90+** while preserving SEO.
- Reduced infrastructure costs by **40%** by replacing cloud-storage-based data delivery with **WebRTC** transfers.
- Built a custom e-commerce platform with **React, Vite, and Shopify**, implementing storefront features and business-specific logic.
- Automated deployment workflows with **GitHub Actions** to standardize the release process.

iStaging

Frontend Engineer | Jan 2022 - Oct 2022

- Built and launched a **Next.js** web platform with SEO support and GA4 product analytics.
- Developed user dashboards and **Three.js**-based VR modules for interactive 3D experiences.

Cyuan You Computer

Frontend Engineer | Jun 2021 - Nov 2021

- Built operational interfaces, merchant management consoles, and corporate portals with **Vue 2** and **Vuetify**.

SELECTED PROJECTS

Phin-Chhù | Interactive Floor Plan Learning App

Live Demo: tw-house-floor-plan.vercel.app

- Built and deployed an interactive learning app for Taiwanese residential floor plans, integrating **symbol search, contextual explanations, and diagram-based quizzes**.
- Designed a **two-step selection flow** with area zoom and candidate confirmation for navigating dense floor plans, alongside on-device learning progress tracking.

TECHNICAL SKILLS

Frontend: React, React Hooks, Next.js, TypeScript, JavaScript, Vite, Three.js

State & Data: Redux, RTK Query, TanStack Query, Zustand

Protocols & Integration: WebRTC, WebSocket, Cross-platform WebView

AI & Developer Productivity: Claude Code, Custom Skills, OpenSpec, Workflow Automation

Delivery: GitHub Actions, CI/CD, AWS

Analytics: Google Analytics 4 (GA4), Google Tag Manager (GTM), Amplitude

EDUCATION

National Yunlin University of Science and Technology | Bachelor of Business Administration (B.B.A.) | 2015 - 2017

National Taichung University of Science and Technology | International Trade and Business | 2010 - 2015